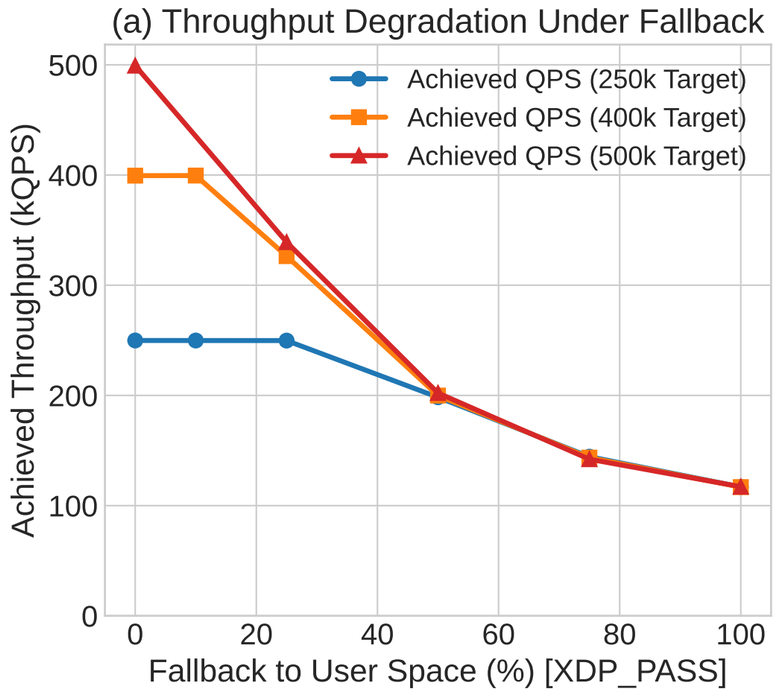
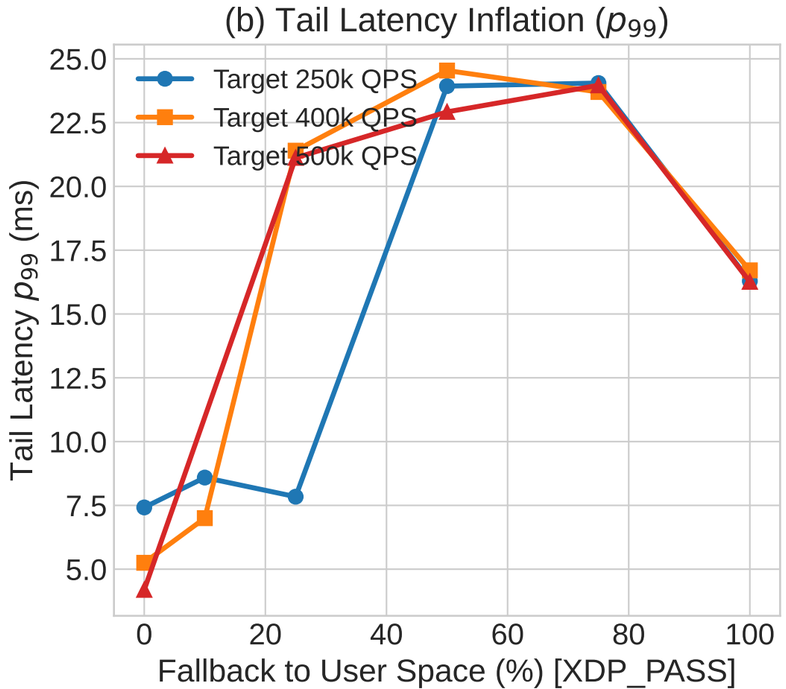
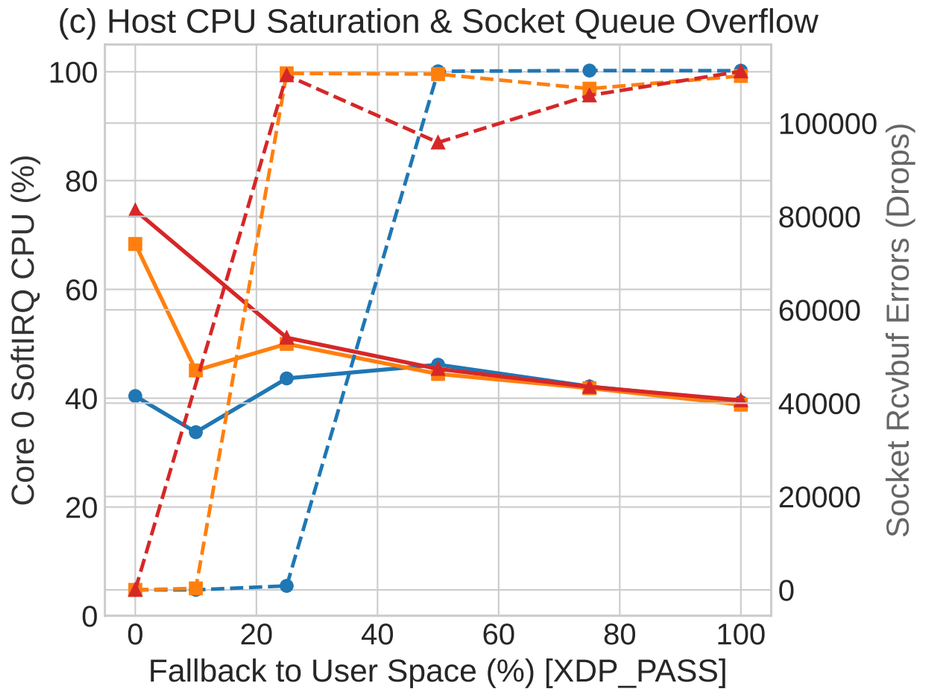

(a) Throughput Degradation Under Fallback
Achieved Throughput (kQPS)
Fallback to User Space (%) [XDP_PASS]
Achieved QPS (250k Target)
Achieved QPS (400k Target)
Achieved QPS (500k Target)
0
100
200
300
400
500
0
20
40
60
80
100


(b) Tail Latency Inflation (p_{99})
Tail Latency p_{99} (ms)
Fallback to User Space (%) [XDP_PASS]
Target 250k QPS
Target 400k QPS
Target 500k QPS
5.0
7.5
10.0
12.5
15.0
17.5
20.0
22.5
25.0
0
20
40
60
80
100


(c) Host CPU Saturation & Socket Queue Overflow
Core 0 SoftIRQ CPU (%)
Socket Rcvbuf Errors (Drops)
Fallback to User Space (%) [XDP_PASS]
0
20
40
60
80
100
0
20000
40000
60000
80000
100000